TAYLAN ÖZGÜR ÖZDEMIR

Software Engineer — Automation, AI & Systems

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EDUCATION

İnönü University *Sep 2021 – Jun 2025*
Bachelor of Science in Computer Engineering — **GPA: 3.09 / 4.00** Malatya, Turkey
Relevant Coursework: Computer Vision, Data Structures & Algorithms, Distributed Systems, Operating Systems, Machine Learning

PROFESSIONAL EXPERIENCE

SoloRota *Jul 2025 – Jan 2026*
Software & Systems Developer (Full-time) Ankara, Turkey (On-site)

- Engineered asynchronous REST API ingestion services and schema validation pipelines for **7 production apps** delivered to live markets.
- Conducted system-level CPU profiling and memory leak tracing, reducing garbage collection spikes by **25%** and ensuring stable throughput.
- Automated build verification pipelines and configuration deployment routines using custom scripts and Git version control workflows.
- Architected decoupled state management layers and event-driven data flow models, accelerating multi-platform build iteration cycles.

KEY PROJECTS

Job-Radar — Autonomous Web Mining & Candidate Scoring Engine *github.com/taylan1477/Job-Radar*
Python, Playwright, Pydantic, Scikit-learn, Google Sheets API, Dynamic DOM Scraping

- Developed an end-to-end web scraping engine using **Playwright** to extract job postings across dynamic Single Page Applications (SPAs) while bypassing bot mitigations.
- Built an NLP-driven evaluation algorithm using **TF-IDF vectorization** and cosine similarity to rank candidates against criteria with **90%+ match accuracy**.
- Integrated automated ingestion pipelines syncing validated data to Google Sheets and Excel via Pydantic data schemas.

Medical CT Image Segmentation — Classical CV vs Deep Learning *github.com/taylan1477*
Python, PyTorch, OpenCV, ScaleNorm-UNet, BRISQUE Quality Assessment, AMOS CT Dataset

- Built a biomedical CT segmentation pipeline benchmarking classical feature extraction (HOG, SIFT, CoHOG, LBP) against **ScaleNorm-UNet**.
- Implemented blind image quality evaluation using **BRISQUE scoring** to filter low-contrast slices, integrated with bilateral and morphological filters.
- Optimized deep learning training runs with custom Dice Loss and IoU metrics, achieving precise boundary delineation on the AMOS CT dataset.

Voice OS (Jarvis) — Offline Autonomous Edge AI Agent *github.com/taylan1477*
Python, Ollama (Qwen/LLaVA), Faster-Whisper, Win32 IPC / OS Automation, Graphify AST

- Engineered an offline edge AI assistant running local quantized models (Qwen 2.5 7B, LLaVA 7B) on local GPU with zero external cloud dependencies.
- Built a low-latency speech pipeline pairing **Faster-Whisper (INT8)** with an interrupt-driven TTS engine (barge-in support) for real-time interaction.
- Integrated OS-level Win32 IPC, background process orchestration, and desktop automation; mapped codebase architecture via deterministic AST knowledge graphs (Graphify).

TECHNICAL SKILLS

AI & Machine Learning: PyTorch, OpenCV, Computer Vision, ScaleNorm-UNet, Scikit-learn, Ollama (Local LLMs), Faster-Whisper, Prompt Engineering

Web Scraping & Systems Automation: Playwright, Selenium, Beautiful Soup, Dynamic DOM Extraction, Anti-Bot Handling, Win32 IPC

Languages & Frameworks: Python, C#, Java, SQL, TypeScript, Pydantic, FastAPI, REST APIs

Systems, Infrastructure & Tools: Docker, Linux (Ubuntu/WSL2), Git, GitHub, Bash, Model Context Protocol (MCP), CI/CD basics